

Multimodal RAG-LLM for Visual QnA in Healthcare

A Dynamic Caption-Cloud + Joint Multimodal-Embedding Retrieval Framework for Clinically-Grounded Visual Answers

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Abstract

We introduce a research prototype and evaluation plan for **Dynamic Multimodal RAG-LLM for Visual QnA**, a retrieval-first multimodal system designed for clinical visual question answering. The core innovation is (A) dynamically generating a dense *caption cloud* for each input image using multiple VLMs (LLaVA-Med, BioMedCLIP, GPT-4V) and diverse prompt templates, and (B) creating a temporary vector database containing *joint multimodal exemplar embeddings* (image + caption) specific to that image. At query time we *clinicalize* the user’s free-text query into a concise third-person medical search string, compute a *joint image+query* multimodal embedding, and retrieve the Top-5 multimodal neighbors from the dynamic database. Each neighbor provides a grounded evidence bundle (caption + provenance) which is used to generate per-neighbor candidate answers. We then synthesize a final clinician-style verdict using principled fusion (centroid embedding fusion, atomic fact intersection, and a meta-LLM scorer), accompanied by an explicit grounding score and provenance mapping. We formalize the model, provide precise mathematical notation, algorithms, and evaluation metrics focused on grounding, calibration, and human clinical judgment, and describe an implementation/reproducibility plan.

1 Motivation and high-level idea

Visual clinical questions are challenging because (1) images are ambiguous, often containing multiple co-occurring findings, (2) users use lay terminology that poorly matches clinical resources, and (3) single captions or single embeddings are brittle. Our approach addresses these by (a) dynamically generating many plausible textual narrations of the input image (caption cloud) using multiple VLMs, and (b) making retrieval itself *multimodal* — we index caption vectors derived from the input image + each generated caption, so retrieval is performed in a joint image+text latent space. This yields neighbors that are intrinsically cross-modal matches to the user’s image and clinical intent, with all evidence directly pertaining to the specific input image.

2 Related work (short)

We build on: (i) vision-language pretraining and instruction tuning for caption generation (BLIP-2, LLaVA, LLaVA-Med); (ii) medical multimodal pretraining (BioMedCLIP, Med-Flamingo) and medical VQA datasets (VQA-RAD, PathVQA, ImageCLEF/VQA-Med); and (iii) retrieval and grounding work (RAG, cross-encoder reranking, atomic fact evaluation). Our novelty is the dynamic caption cloud generation + storing *multimodal* caption vectors as the primary index key and fusing Top-5 evidence-conditioned answers, all specific to the input image.

3 Problem statement and notation

Let:

- I be the input image (user supplied).

- q_{user} be the user question (free text).
- $g(\cdot)$ be the clinicalizer that rewrites $q_{\text{user}} \mapsto q_{\text{clin}}$ (short, clinical string).
- $\mathcal{M} = \{M_1, M_2, \dots, M_V\}$ be the set of VLMs (LLaVA-Med, BioMedCLIP, GPT-4V, etc.).
- $\Pi = \{\pi_1, \pi_2, \dots, \pi_P\}$ be the set of prompt templates.
- $\mathcal{C}_{\text{dyn}}^{(I)} = \{c_i\}_{i=1}^C$ be the dynamic caption cloud generated for input image I .
- $E_v : \mathcal{I} \rightarrow \mathbb{R}^{d_v}$ be an image encoder, $E_t : \mathcal{T} \rightarrow \mathbb{R}^{d_t}$ be a text encoder.
- $F_m : \mathbb{R}^{d_v} \times \mathbb{R}^{d_t} \rightarrow \mathbb{R}^d$ be a multimodal fusion encoder producing joint vectors.
- The dynamic index stores tuples $(\phi_i, c_i, \text{meta}_i)$ where $\phi_i = F_m(W_v E_v(I), W_t E_t(c_i))$.

Our goal: given (I, q_{user}) produce a final answer A with (i) explicit provenance linking claims to caption IDs, and (ii) grounding score $g(A) \in [0, 1]$ reflecting evidence support from the dynamic caption cloud.

4 Method — detailed

We describe the dynamic caption generation and the online query pipeline in detail, with formulas and reasoning.

4.1 Dynamic caption cloud generation

For each input image I , generate a dynamic set of captions:

$$\mathcal{C}_{\text{dyn}}^{(I)} = \bigcup_{j=1}^V \bigcup_{k=1}^P \bigcup_{s=1}^S \{c_{j,k,s}\}$$

by running VLM M_j with prompt template π_k and sampling seed s :

$$c_{j,k,s} = M_j(I; \pi_k, s).$$

VLM Ensemble ($\mathcal{M}$):

- **LLaVA-Med**: Medical domain fine-tuned vision-language model
- **BioMedCLIP**: Biomedical vision-language pretraining
- **GPT-4V**: General multimodal with strong reasoning
- **Med-Flamingo**: Medical instruction-following model
- **CheXzero**: Chest X-ray specialist (when applicable)

Prompt Template Bank (II):

- Clinical-structured: “Describe the location, morphology, distribution, and associated features”
- Attribute-focused: “What are the color, texture, and surface characteristics?”
- Question→caption: “Is there evidence of inflammation? Describe what you observe”
- Differential-focused: “What are the key diagnostic features for differential diagnosis?”
- Triage-oriented: “What immediate concerns or red flags are visible?”

Retain provenance (model, prompt, seed) for each caption. Deduplicate by fuzzy similarity but preserve paraphrase diversity (keep semantic variants).

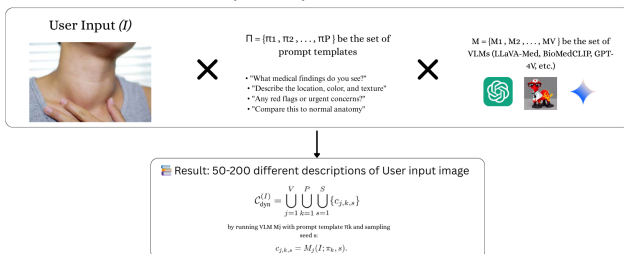


Fig. 1: Caption Cloud Generation

Rationale: A dense dynamic cloud increases support coverage for many possible clinical phrasings specific to the input image and reduces risk that a single brittle caption misses relevant evidence.

4.2 Dynamic multimodal embedding and temporary index

For the input image I with its dynamic caption cloud $\mathcal{C}_{\text{dyn}}^{(I)}$, compute the image embedding once:

$$z^{(I)} = E_v(I) \in \mathbb{R}^{d_v} \quad (1)$$

For each caption $c_i \in \mathcal{C}_{\text{dyn}}^{(I)}$, compute:

$$\tau_i = E_t(c_i) \in \mathbb{R}^{d_t} \quad (2)$$

Project and fuse (learned linear projectors $W_v \in \mathbb{R}^{d \times d_v}$, $W_t \in \mathbb{R}^{d \times d_t}$):

$$\phi_i = F_m(W_v z^{(I)}, W_t \tau_i) \in \mathbb{R}^d \quad (3)$$

Store in temporary vector database:

$$\text{DB}_{\text{temp}}^{(I)} = \{(\phi_i, c_i, \text{meta}_i)\}_{i=1}^{|\mathcal{C}_{\text{dyn}}^{(I)}|}$$

where meta_i contains (M_j, π_k, s) provenance information.

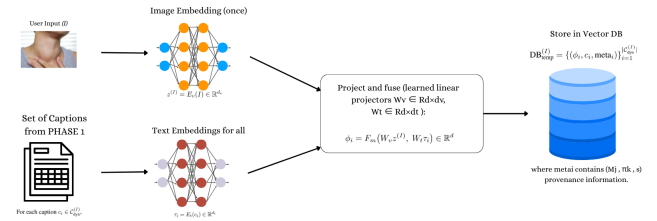


Fig. 2: Retrieval Database creation

Key Innovation: Unlike traditional RAG systems with pre-built indices, we create a personalized vector database containing only captions of the specific input image, ensuring all retrieved evidence directly pertains to the user’s query image.

4.3 Query clinicalization and joint query embedding

Given q_{user} , compute clinical rewrite:

$$q_{\text{clin}} = g(q_{\text{user}}),$$

using a constrained prompt to an LLM or a hybrid LLM + entity normalizer (scispaCy/QuickUMLS). Then compute:

$$z_q = E_v(I), \quad \tau_q = E_t(q_{\text{clin}}),$$

and the query multimodal key:

$$\phi_q = F_m(W_v z_q, W_t \tau_q). \quad (4)$$

Key design decision: this ϕ_q is the canonical retrieval key and is used to query the temporary index of ϕ_i vectors. That is, retrieval is fully multimodal (image + clinical text) performed in a joint latent space against the dynamic caption database.

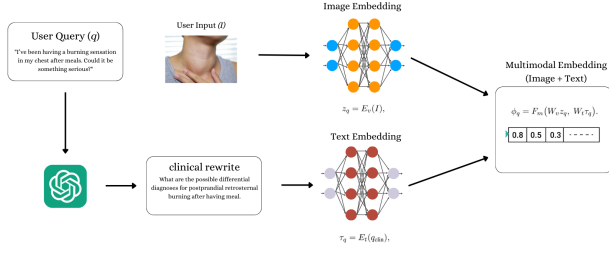


Fig. 3: Query Clinicalization, Image Embedding and fusion

4.4 Multimodal retrieval from dynamic database (Top-K, K=5)

Query the temporary index with ϕ_q and retrieve Top- K nearest neighbors under cosine similarity:

$$\mathcal{R} = \text{TopK}_{\phi}(\phi_q, \text{DB}_{\text{temp}}^{(I)}; K = 5) = \{(\phi_n, s_n, c_n, \text{meta}_n)\}_{n=1}^5,$$

where $s_n = \cos(\phi_q, \phi_n)$.

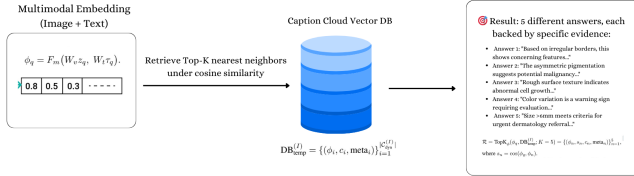


Fig. 4: Retrieval from Database

Interpretation: each neighbor is a caption of the input image whose joint semantics (when paired with the image) match both the user’s visual content and the clinicalized query intent.

4.5 Per-neighbor evidence conditioning and candidate generation

For each neighbor $n \in \mathcal{R}$ we assemble evidence bundle $E_n = \{c_n, s_n, \text{meta}_n\}$. We then produce a candidate answer via a constrained generator G :

$$a_n, p_n, \Gamma_n = G(I, q_{\text{clin}}, E_n), \quad (5)$$

where:

- a_n is the generated textual answer conditioned on the query, original image and evidence bundle,
- p_n is a generator confidence proxy (normalized token log probability or calibration head),
- Γ_n is a per-claim grounding map that links atomic claims in a_n to the retrieved caption c_n (enforced by prompt constraints).

4.6 Fusion strategies for final verdict

We now have candidate set $\mathcal{A} = \{(a_n, p_n, s_n, \Gamma_n)\}_{n=1}^5$. We present three principled fusion operators and their math:

1. Centroid embedding fusion (fluency + weighted evidence) Embed answers: $u_n = E_t(a_n) \in \mathbb{R}^{d_t}$. Define weights incorporating VLM diversity:

$$w_n = \gamma_1 p_n + \gamma_2 s_n + \gamma_3 \cdot \text{diversity}(\text{meta}_n),$$

where $\text{diversity}(\text{meta}_n)$ rewards captions from different VLMs or prompt templates. Compute centroid:

$$u^* = \frac{\sum_{n=1}^5 w_n u_n}{\sum_{n=1}^5 w_n}.$$

Produce final a^* by either (a) selecting the a_n nearest to u^* or (b) summarizing u^* via a paraphrase generator constrained to cite supporting caption sources.

2. Atomic-fact intersection (high-precision) Decompose each a_n into atomic facts $F(a_n) = \{f_{n,k}\}$. Let support count:

$$\text{count}(f) = |\{n : f \in F(a_n)\}|.$$

Retain facts with $\text{count}(f) \geq m$ (e.g., majority $m = 3$). Assemble a^* by ordering retained facts by support and attaching provenance (VLM and prompt source). This method prioritizes precision and reduces hallucinated facts.

3. Meta-LLM scoring (flexible synthesis) Provide a meta-LLM with the full tuple $\{(a_n, p_n, s_n, \Gamma_n, \text{meta}_n)\}$ and ask it to synthesize a^* and a final grounding score g^* . Constrain output to a JSON schema including evidence lists with VLM provenance.

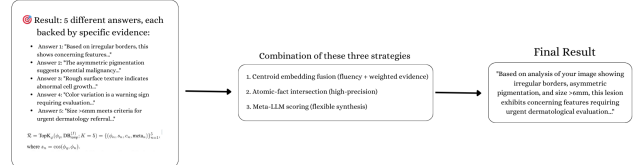


Fig. 5: Fusion strategy

These fusion strategies can be combined: run intersection to get high-precision core claims, run centroid to create readable narrative around them, and use meta-LLM to produce the final clinically palatable answer.

4.7 Final grounding score and refusal

Define diagnostics incorporating the dynamic nature:

$$\bar{s} = \frac{1}{5} \sum_n s_n, \quad \bar{p} = \frac{1}{5} \sum_n p_n,$$

$$\text{consensus} = \frac{\sum_f \mathbb{1}\{\text{count}(f) \geq m\}}{\sum_f \mathbb{1}\{\exists n : f \in F(a_n)\}}.$$

VLM diversity score:

$$\text{diversity} = \frac{|\{\text{VLM sources in } \mathcal{R}\}|}{|\mathcal{M}|}.$$

Then compute:

$$g^* = \eta_1 \bar{s} + \eta_2 \bar{p} + \eta_3 \cdot \text{consensus} + \eta_4 \cdot \text{diversity}, \quad (6)$$

where weights η are calibrated on held-out clinician judgments (optimize ECE). If $g^* < \tau_{\text{refuse}}$ the system must refuse diagnostic claims and only produce descriptive observations + triage steps.

5 The multimodal encoder F_m — detailed design

We detail three practical instantiations (in increasing order of modeling power and engineering cost):

5.1 Concatenate + MLP (baseline)

Project $z \leftarrow W_v E_v(I)$ and $\tau \leftarrow W_t E_t(t)$ then:

$$F_m(z, \tau) = \text{LayerNorm}(\text{MLP}([z; \tau])).$$

Pros: simple, fast; cons: limited cross-modal interaction modeling.

5.2 Cross-attention transformer (recommended)

Tokenize image features into visual tokens (patch features) $V \in \mathbb{R}^{T_v \times d'}$, tokenize text into tokens $T \in \mathbb{R}^{T_t \times d'}$, then apply cross-attention layers where text attends to vision and vice versa. Pool output:

$$F_m(V, T) = \text{Pool}(\text{Transformer}_{\text{cross}}(V, T)).$$

This captures fine spatial references (e.g., “redness adjacent to the conjunctival margin”).

5.3 Contrastive pretraining + fusion adapter

Train/fine-tune a CLIP-style contrastive model on image+caption pairs in the clinical corpus so that E_v and E_t are aligned; then apply a small MLP fusion on aligned vectors. This is scalable and aligns modalities before fusion.

6 Re-ranking and diversity

After retrieving Top-K neighbors from the dynamic database, compute a re-ranking score $\rho_n = r(q_{\text{clin}}, c_n)$ using a cross-encoder or small LLM scoring head; re-rank candidates, cluster by semantic similarity and pick cluster representatives to ensure the evidence set is diverse (avoid near-duplicates from similar VLM outputs).

7 Implementation details and engineering

Dynamic indexing Use FAISS for fast in-memory similarity search with no persistence needed. For each query, build index from scratch with $|C_{\text{dyn}}^{(I)}|$ vectors (typically 50-200 captions).

Caption cloud scale Prototype: $C \approx 50 - 200$ captions / image using 3-5 VLMs with 5-10 prompt templates and 2-3 seeds each. Cache generated captions during processing to avoid redundant VLM calls.

VLM infrastructure

- API integration for GPT-4V, Claude Vision
- Local deployment for LLaVA-Med, BioMedCLIP via transformers/open_clip
- Batch processing for efficiency
- Timeout and error handling for robust operation

Embeddings Use robust text embedder (bge-m3 / high-quality SBERT) for E_t ; use domain adapted image encoders (BioMedCLIP or a fine-tuned ViT) for E_v . Where possible, adapt E_v/E_t contrastively on in-domain data.

Generator Use a constrained instruction-tuned LLM (or multimodal LLM) for G and the meta-LLM; enforce strict JSON schema and prompt constraints to ensure claims are linked to caption sources with VLM provenance.

8 Evaluation protocol — rigorous

8.1 Datasets

Use VQA-RAD, VQA-Med/ImageCLEF, PathVQA for benchmarks plus a curated clinical testbed (200–500 clinician-annotated skin/eye images with gold atomic facts and triage labels).

8.2 Metrics

- Standard: VQA accuracy, EM, BLEU/ROUGE for open answers.
- **Grounding Precision (GP)**: fraction of atomic claims in a^* that have supporting evidence in retrieved captions from $\mathcal{R}$.
- **Caption Relevance (CR)**: relevance of retrieved captions to the clinical query (human evaluation).
- **VLM Consensus (VC)**: agreement among different VLMs for similar findings.
- **Dynamic Latency (DL)**: end-to-end response time including caption generation.
- **Calibration (ECE)**: of g^* vs empirical correctness.
- **Human clinician study**: rate usefulness, trust, and safety; measure clinician preference vs static RAG systems.

9 Safety, bias and ethics

Practical policies:

- Strip PII/EXIF; provide explicit consent screens for users.
- Disallow treatment advice when $g^* < \tau_{\text{refuse}}$; always provide triage pathways.
- Log VLM provenance and create clinician audit interface for flagged cases.
- Monitor VLM bias across demographic groups; implement fairness constraints.
- Maintain dataset licensing and patient consent; do not store user images beyond session.

10 Failure modes and mitigation

1. **VLM hallucination:** mitigate by intersection fusion, requiring multiple VLM agreement, and confidence thresholding.
2. **Wrong clinicalization:** keep original q_{user} available to the generator and log rewrites; build human-in-loop correction.
3. **Caption redundancy:** implement semantic deduplication while preserving useful paraphrases.
4. **Computational cost:** optimize VLM batching, implement early stopping, cache common patterns.

11 Conclusion

This proposal presents a rigorous, reproducible research prototype where retrieval is executed against a dynamically generated, image-specific caption database in a joint image+text latent space. By generating multiple captions from diverse VLMs for each input image, then retrieving and synthesizing evidence-conditioned candidate answers with explicit provenance and calibration, we achieve a more interpretable and grounded Visual QnA system that ensures all evidence directly pertains to the user’s specific image, making it well-suited to clinical decision support research.

A Example prompts and constrained JSON schema

Clinical Structured Prompt: “Provide a detailed medical description of this image, including: 1) Anatomical location and orientation, 2) Morphological characteristics of any findings, 3) Color, texture, and surface properties, 4) Distribution pattern and symmetry, 5) Associated features or secondary findings.”

Triage-Oriented Prompt: “From a clinical triage perspective, identify: 1) Any immediate red flags or concerning

features, 2) Findings that suggest urgent vs routine care, 3) Key diagnostic clues visible in this image, 4) What additional information would be most helpful for diagnosis.”

B Algorithm (pseudocode)

Algorithm 1 Dynamic Multimodal RAG-LLM

```

1: Input: image  $I$ , user query  $q_{\text{user}}$ 
2: // Dynamic caption generation
3:  $\mathcal{C}_{\text{dyn}}^{(I)} \leftarrow \emptyset$ 
4: for each VLM  $M_j \in \mathcal{M}$  do
5:   for each prompt  $\pi_k \in \Pi$  do
6:     for  $s = 1$  to  $S$  do
7:        $c \leftarrow M_j(I, \pi_k, s)$ 
8:        $\mathcal{C}_{\text{dyn}}^{(I)} \leftarrow \mathcal{C}_{\text{dyn}}^{(I)} \cup \{(c, M_j, \pi_k, s)\}$ 
9:     end for
10:  end for
11: end for
12: // Create temporary vector database
13:  $z^{(I)} \leftarrow E_v(I)$ 
14:  $\text{DB}_{\text{temp}}^{(I)} \leftarrow \emptyset$ 
15: for each  $(c_i, \text{meta}_i) \in \mathcal{C}_{\text{dyn}}^{(I)}$  do
16:    $\tau_i \leftarrow E_t(c_i)$ 
17:    $\phi_i \leftarrow F_m(W_v z^{(I)}, W_t \tau_i)$ 
18:    $\text{DB}_{\text{temp}}^{(I)} \leftarrow \text{DB}_{\text{temp}}^{(I)} \cup \{(\phi_i, c_i, \text{meta}_i)\}$ 
19: end for
20: // Query processing and retrieval
21:  $q_{\text{clin}} \leftarrow g(q_{\text{user}})$ 
22:  $\tau_q \leftarrow E_t(q_{\text{clin}})$ 
23:  $\phi_q \leftarrow F_m(W_v z^{(I)}, W_t \tau_q)$ 
24:  $\mathcal{R} \leftarrow \text{TopK}_{\phi}(\phi_q, \text{DB}_{\text{temp}}^{(I)}, K = 5)$ 
25: for each  $(\phi_n, c_n, \text{meta}_n) \in \mathcal{R}$  do
26:    $E_n \leftarrow (c_n, \cos(\phi_q, \phi_n), \text{meta}_n)$ 
27:    $(a_n, p_n, \Gamma_n) \leftarrow G(I, q_{\text{clin}}, E_n)$ 
28: end for
29:  $a^* \leftarrow \mathcal{F}(\{a_n, p_n, s_n, \Gamma_n\})$  ▷ fusion
30: compute  $g^*$  with VLM diversity; if  $g^* < \tau_{\text{refuse}}$  restrict output
31: return  $a^*$ , evidence list with VLM provenance,  $g^*$ 

```
